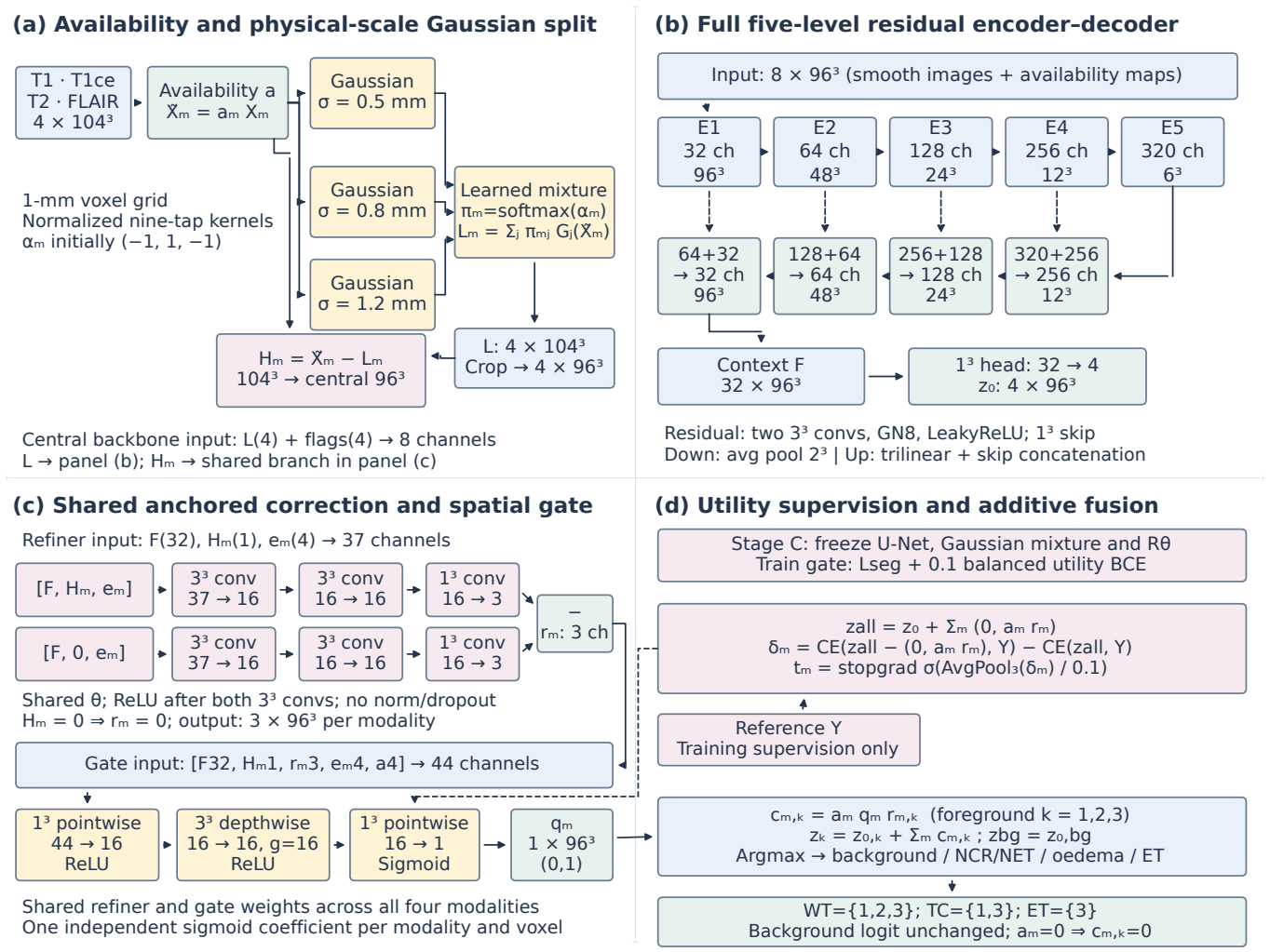
(a) Availability and physical-scale Gaussian split
T1 · T1ce
T2 · FLAIR
4 × 104³
Availability a
X̃m = am Xm
Gaussian
σ = 0.5 mm
Gaussian
σ = 0.8 mm
Gaussian
σ = 1.2 mm
Learned mixture
πm=softmax(αm)
Lm = Σj πmj Gj(X̃m)
1-mm voxel grid
Normalized nine-tap kernels
αm initially (−1, 1, −1)
L: 4 × 104³
Crop → 4 × 96³
Hm = X̃m − Lm
104³ → central 96³
Central backbone input: L(4) + flags(4) → 8 channels
L → panel (b); Hm → shared branch in panel (c)
(b) Full five-level residual encoder–decoder
Input: 8 × 96³ (smooth images + availability maps)
E1
32 ch
96³
E2
64 ch
48³
E3
128 ch
24³
E4
256 ch
12³
E5
320 ch
6³
64+32
→ 32 ch
96³
128+64
→ 64 ch
48³
256+128
→ 128 ch
24³
320+256
→ 256 ch
12³
Context F
32 × 96³
1³ head: 32 → 4
z0: 4 × 96³
Residual: two 3³ convs, GN8, LeakyReLU; 1³ skip
Down: avg pool 2³ | Up: trilinear + skip concatenation
(c) Shared anchored correction and spatial gate
Refiner input: F(32), Hm(1), em(4) → 37 channels
[F, Hm, em]
3³ conv
37 → 16
3³ conv
16 → 16
1³ conv
16 → 3
[F, 0, em]
3³ conv
37 → 16
3³ conv
16 → 16
1³ conv
16 → 3
rm: 3 ch
Shared θ; ReLU after both 3³ convs; no norm/dropout
Hm = 0 ⇒ rm = 0; output: 3 × 96³ per modality
Gate input: [F32, Hm1, rm3, em4, a4] → 44 channels
1³ pointwise
44 → 16
ReLU
3³ depthwise
16 → 16, g=16
ReLU
1³ pointwise
16 → 1
Sigmoid
qm
1 × 96³
(0,1)
Shared refiner and gate weights across all four modalities
One independent sigmoid coefficient per modality and voxel
(d) Utility supervision and additive fusion
Stage C: freeze U-Net, Gaussian mixture and Rθ
Train gate: Lseg + 0.1 balanced utility BCE
zall = z0 + Σm (0, am rm)
δm = CE(zall − (0, am rm), Y) − CE(zall, Y)
tm = stopgrad σ(AvgPool3(δm) / 0.1)
Reference Y
Training supervision only
cm,k = am qm rm,k (foreground k = 1,2,3)
zk = z0,k + Σm cm,k ; zbg = z0,bg
Argmax → background / NCR/NET / oedema / ET
WT={1,2,3}; TC={1,3}; ET={3}
Background logit unchanged; am=0 ⇒ cm,k=0